What changes when the model changes?

A freeze-drying cycle must finish primary drying without exceeding product-temperature or equipment-capacity limits. This experiment compares forecast errors and proposed operating choices when an AI agent receives different physical tools.

Protocol scope: Post-hoc motivated, prospectively frozen same-case subset; joint budget and submission-policy change; not independent confirmation. Recorded completion policy: Final eighth provider turn exposes only submit_decision with forced function choice, identically across all three arms; earlier submission allowed; eight host calls include submission.

The engineering change

Model B represents one quasi-steady vial. Model C represents explicit weighted vial populations with independent ice mass and thermal storage. C tracks the hottest still-drying population; already-dry vials can warm without becoming active primary-drying violations.

Both models retain prescribed chamber pressure. Both use the same rule: once predicted capacity is exceeded, completion is not a valid physical forecast. The change does not add a pressure-dynamics model or establish physical validation.

Two distinct decisions

Forecast: use early synthetic receiving observations to predict the known KEEP schedule and choose among the allowed actions. Pretransfer: make the same forecasts and choices from characterization alone, without receiving observations. The tasks are never combined into one accuracy number.

What was compared

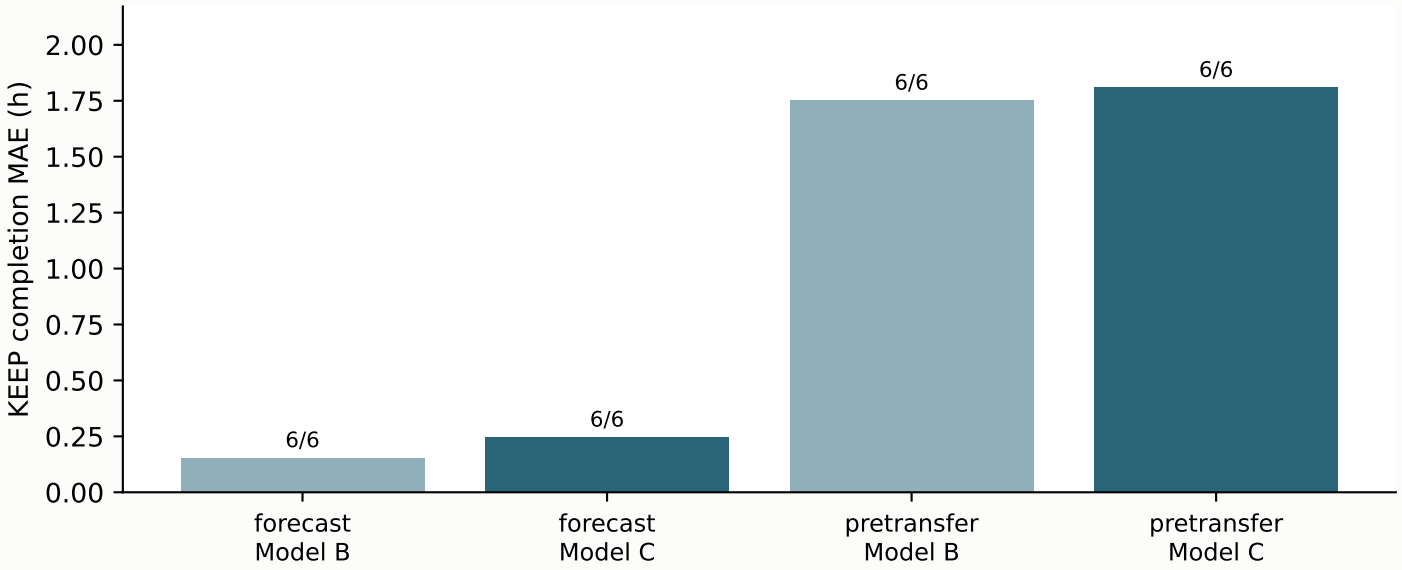
Deterministic B/C controls select from model predictions. AI A receives no prebuilt physical model and can derive its own simulator using Python. AI B and AI C receive prepared legacy and population tools. The candidate action grid also permits characterize.

All measurements and reference outcomes are synthetic. Reference completion and subsequent thermal outcomes are censored after capacity exceedance. The evaluator and candidates share constitutive laws; this is an exploratory comparison under synthetic mismatch.

Model-only results, with coverage

MAE describes KEEP completion on eligible cases with a numerical prediction. Pred./eligible exposes that denominator; feasible/unsafe describes selected actions. Missing or failed attempts remain in the planned-case score.

Task	Arm	Done / tried	Pred. / eligible	MAE (hours)	Feasible / unsafe	Score* (vials/h)
forecast	Model B	6/6	6/6	0.15	6/0	206.2
forecast	Model C	6/6	6/6	0.25	6/0	198.9
pretransfer	Model B	6/6	6/6	1.75	5/1	178.5
pretransfer	Model C	6/6	6/6	1.81	5/1	179.0



Forecast common-prediction comparison: C minus B MAE = 0.09 h; n = 6. Negative means lower completion error on that common subset.

Pretransfer common-prediction comparison: C minus B MAE = 0.06 h; n = 6. Negative means lower completion error on that common subset.

* Benchmark throughput score: mean synthetic feasible vials/h over planned cases, with zero for invalid choices, characterize, failure or no attempt. This is not real plant throughput.

AI decisions versus prepared controls

Per attempt: 8 model calls, 240 seconds, 8 tool calls. A can build a simulator in Python; B/C receive prepared models. The comparison includes the workflow cost of doing that within these limits.

Completion policy: Final eighth provider turn exposes only submit_decision with forced function choice, identically across all three arms; earlier submission allowed; eight host calls include submission.

Forecast

Arm	Done / tried	Pred. / eligible	MAE (hours)	Feasible / unsafe	Score* (vials/h)
Model B	6/6	6/6	0.15	6/0	206.2
Model C	6/6	6/6	0.25	6/0	198.9
AI A	4/6	4/6	53.99	3/0	47.0
AI B	6/6	6/6	0.16	6/0	202.3
AI C	6/6	6/6	0.25	6/0	192.7

Pretransfer

Arm	Done / tried	Pred. / eligible	MAE (hours)	Feasible / unsafe	Score* (vials/h)
Model B	6/6	6/6	1.75	5/1	178.5
Model C	6/6	6/6	1.81	5/1	179.0
AI A	4/6	4/6	9.05	3/1	114.3
AI B	6/6	6/6	1.75	5/1	164.4
AI C	6/6	6/6	1.81	5/1	178.8

Recorded AI call-budget exhaustions: 2; characterize submissions: 1 across the two tasks. Failures and abstentions remain visible and earn zero benchmark throughput credit. Any prepared-tool workflow advantage is conditional on the fixed budget.

Lower completion MAE does not establish an action-selection gain. Compare coverage, feasible and unsafe selections, and benchmark score separately. These outcomes do not measure human engineering effort or actual manufacturing gains.

Done/tried = completed/attempted; Pred./eligible = numerical KEEP forecasts/reference-eligible cases. Score* is the benchmark throughput score defined on page 2.

What this evidence can and cannot say

Warning support, not a broad safety claim

Forecast reference support among 6 planned cases: 0 positive KEEP capacity outcomes; 1 positive KEEP thermal outcomes. With zero KEEP capacity positives, capacity sensitivity cannot be estimated.

Pretransfer reference support among 6 planned cases: 0 positive KEEP capacity outcomes; 1 positive KEEP thermal outcomes. With zero KEEP capacity positives, capacity sensitivity cannot be estimated.

Important limits

The half-hourly temperature and ice-mass observations are synthetic information channels, not a claim about routine factory sensor availability. Pretransfer receiving shifts are hidden and cannot be recovered from the supplied characterization alone.

The evaluator and candidates share laws and approximate population structure. Thermal nodes are not spatial temperature fields. Post-overload pressure behavior is not a physical forecast target. The tested action grid is limited; characterize can be a reasonable choice despite earning no score.

Next evidence needed

Define the intended decision and acceptable error; independently characterize vial populations, resistance, thermal storage and equipment capacity; then compare predictions with independent experimental or plant data. Include positive capacity cases and a practical observation plan before drawing conclusions about warning sensitivity.

Reproducibility and review

The companion report, per-case CSV and summary JSON preserve counts, missing attempts, source hashes, cost uncertainty and descriptive paired comparisons. No manufacturing actions were executed. This brief is for research review and discussion.

Frozen protocol: https://github.com/YalmanchiliTejas/BioDemo/issues/1